

SUBSTITUTION MODELS

WEEK 37 PHYLOGENETIC TREES



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EVOLUTIONARY THINKING
29. AUGUST 2025

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WHAT ARE WE DOING TODAY

Recap of different phylogenetic trees

Exercises on testing substitution models and making trees



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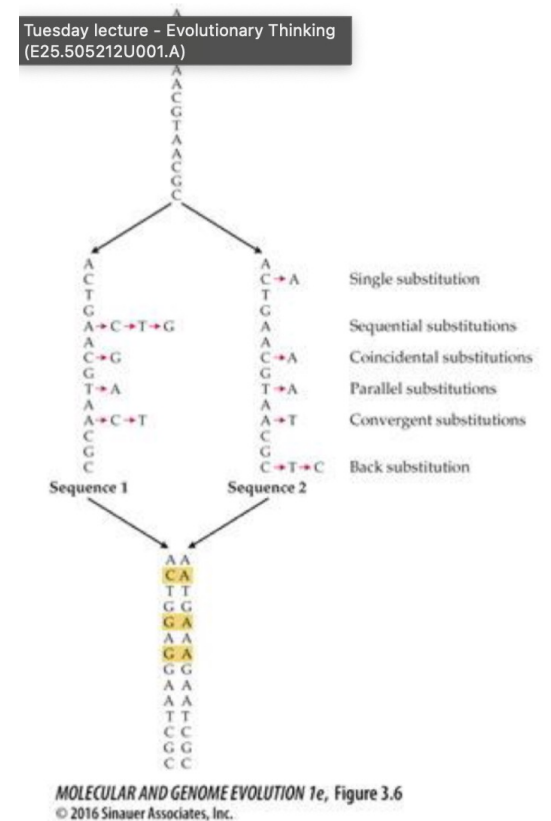


NUCLEOTIDE SUBSTITUTIONS

Divergent sequences

AGATGCTAGCATCGACTAGCARCAGCTGACCCGCGCGCGCAT
AGAT**AA**TAGCATCGACTAGCARCAG**C**GGACCCGCGCGCGCAC**C**

Different substitution models to explain this



SUBSTITUTION MODELS

D-estimation -> proportion of nucleotide sites that differ

Jukes-Cantor -> corrects for multiple & reverse changes

+ separate rates for **transition** ($A \leftrightarrow G$ or $C \leftrightarrow T$) and **transversions**

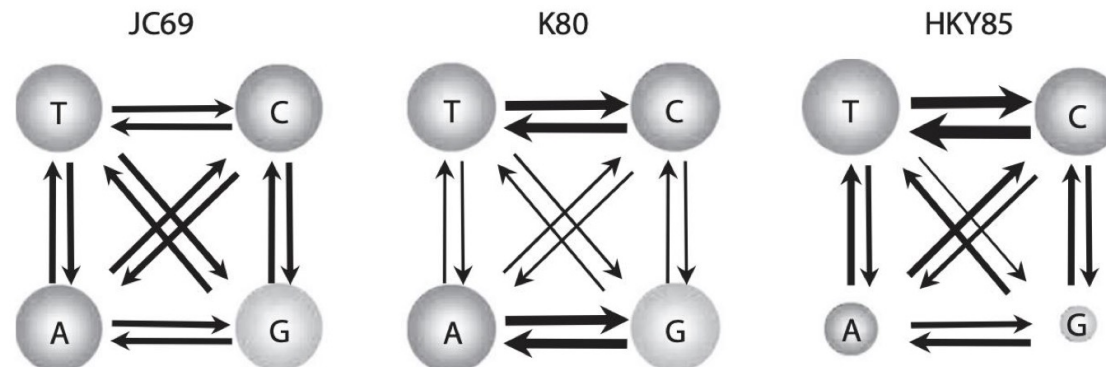
Kimura's 2 parameter model

+ Different equilibrium **frequency** distribution

HKY85

+ **variability** among sites

Model + Gamma

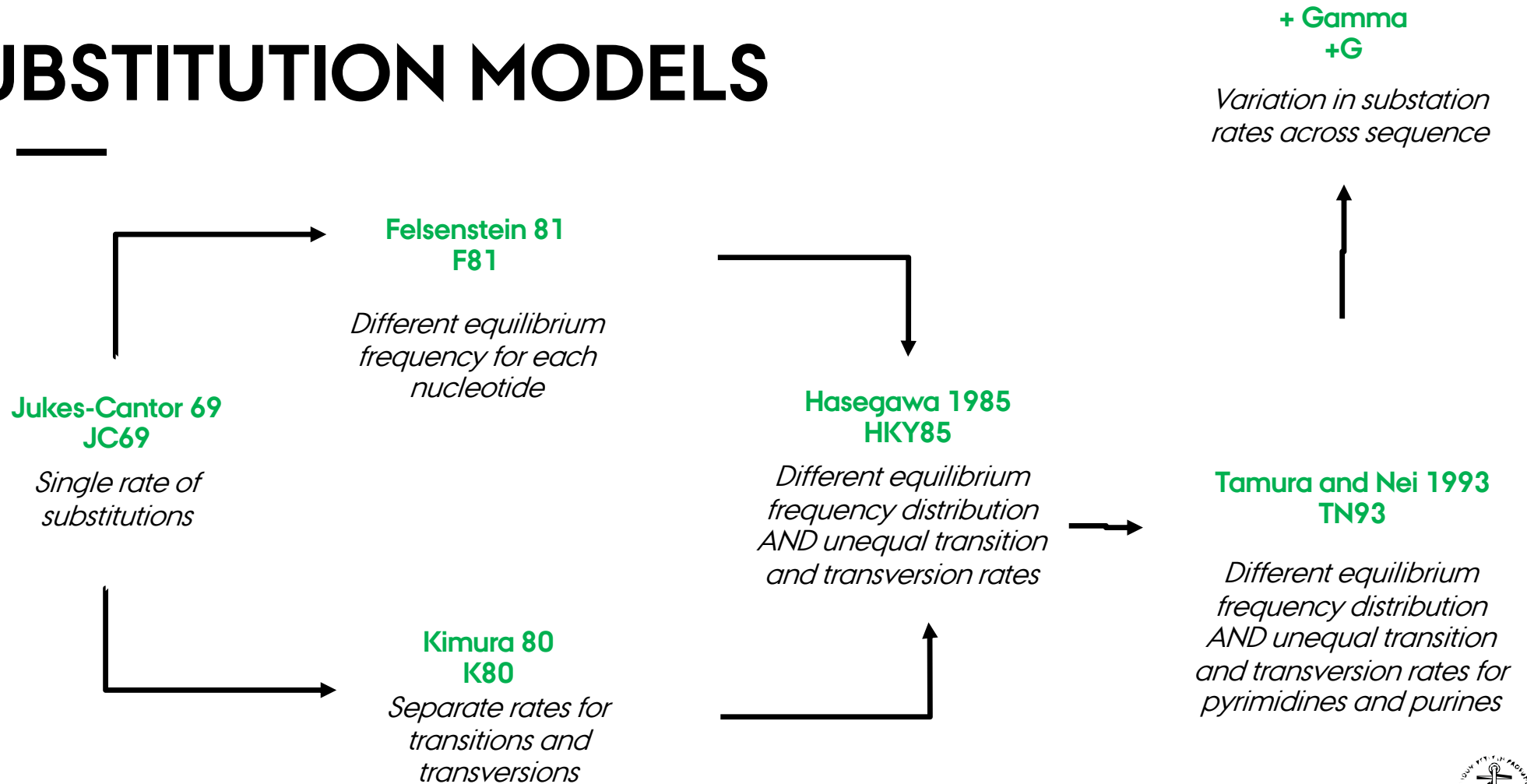


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SUBSTITUTION MODELS



DIFFERENT METHODS – PHYLOGENETIC TREES

Distance method -> Fit a tree whose path-lengths reproduce the distance matrix as closely as possible

- UPGMA
- Neighbor joining

Parsimony method -> Find the tree needing the **fewest total mutations** to explain the actual sequences

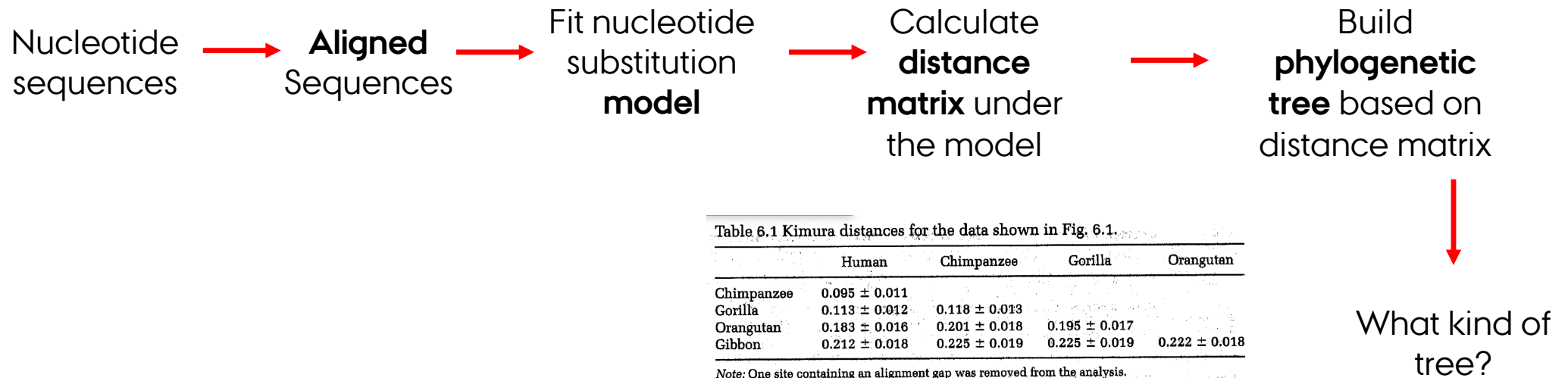
- Maximum parsimony

Likelihood method -> Find the tree (+ branch lengths) that makes the observed sequences **most probable** under the substitution model

- Maximum Likelihood



ORDER TO BUILD PHYLOGENETIC TREE



UPGMA

Repeatedly join the 2 closest things, replace them with a single code and recompute the distance to everything else as an average.

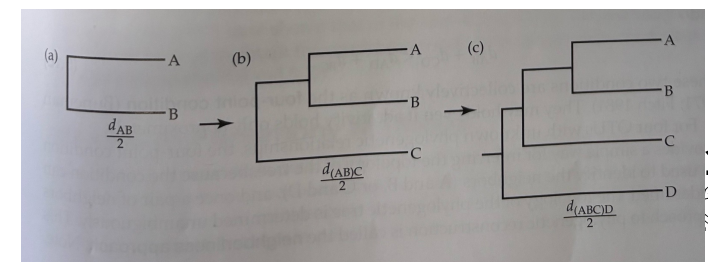
Assumption: evolution rates are constant (= molecular clock)

Sequentially clustering

- Start with the most similar ones (least distant) → they are now one unit
- Recalculate distance matrix
- Add the next most similar one
- Repeat until you are only left with 2 units

OTU	A	B	C
B	d_{AB}		
C	d_{AC}	d_{BC}	
D	d_{AD}	d_{BD}	d_{CD}

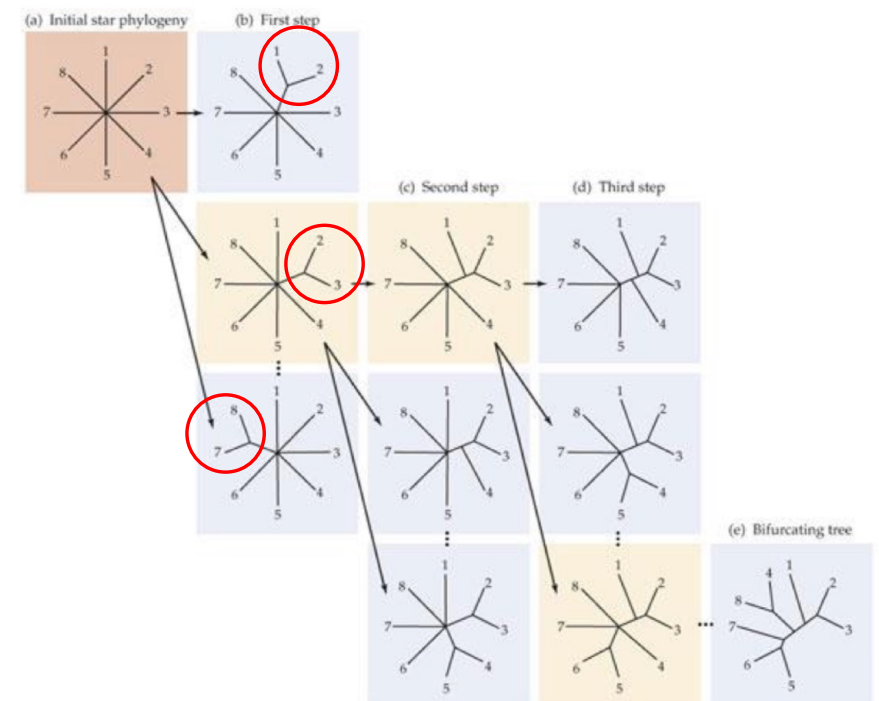
OTU	(AB)	C
C	$d_{(AB)C}$	
D	$d_{(AB)D}$	d_{CD}



NEIGHBOR-JOINING METHOD

Finds the shortest (minimum evolution) tree → sequentially finding neighbors that **minimize the total length of the tree**.

- Start with star → you add a branching and see how the total tree length is still the shortest.
- Continued until all the internal branches are found.



MOLECULAR AND GENOME EVOLUTION 1e, Figure 5.20
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MAXIMUM PARSIMONY

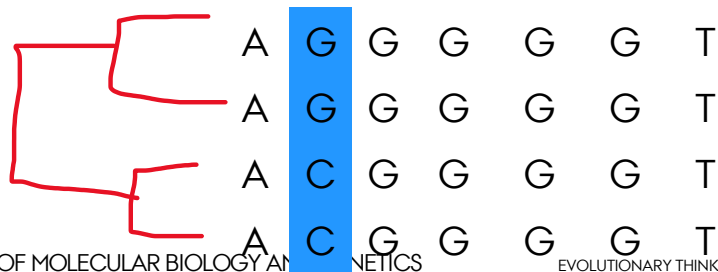
NOT a distance method

Ignore branch length and distances entirely. -> look at the raw sequence

Identification of a topology that requires **the smallest number of evolutionary changes** (= nucleotide substitutions) to explain the observed differences.

= shortest set of pathways leading to the final tree.

You only look at variable sites



MAXIMUM LIKELIHOOD

Using the fitted substitution model → given this tree topology and these branch length, what is the probability of observing exactly the sequence we have?

→ then you search the topologies and branch lengths to find the combination that maximized the probability.

This uses the substitution models -> correctly accounts for multiple/hidden substitutions and other assumptions such as transition/transversion rates, nucleotide frequencies etc.



TODAY

Exercises in MEGA in your group

Try different phylogenetic tree building methods.

Some models do not exist in MEGA or you can get a most likely model but you cannot use that model for the building of a tree → then just pick the next most likely one.



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FRIDAY – WORK ON ASSIGNMENT

Work on hand-in assignment I

you make this in your group



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